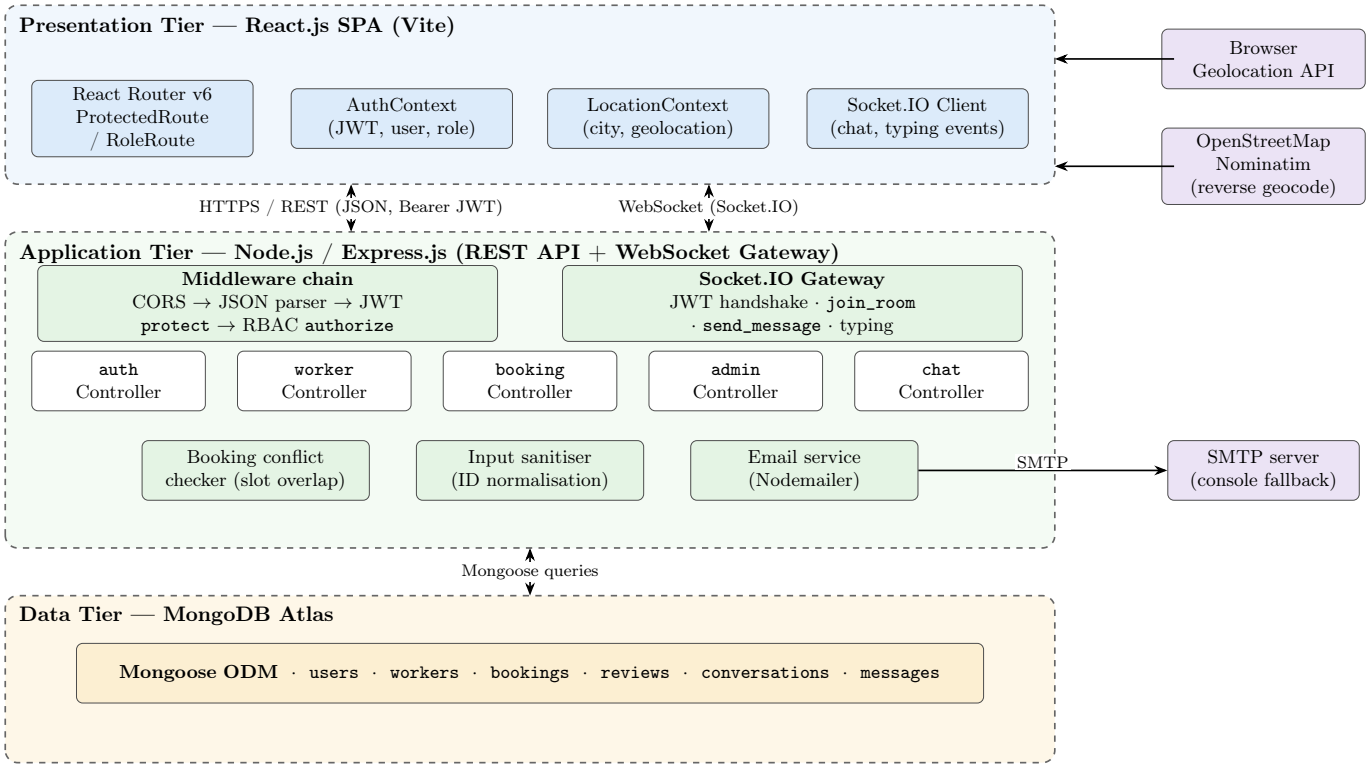


Figure 1: High-level system architecture of ServeConnect (three-tier, MERN-based, with a WebSocket gateway and SMTP notification service).

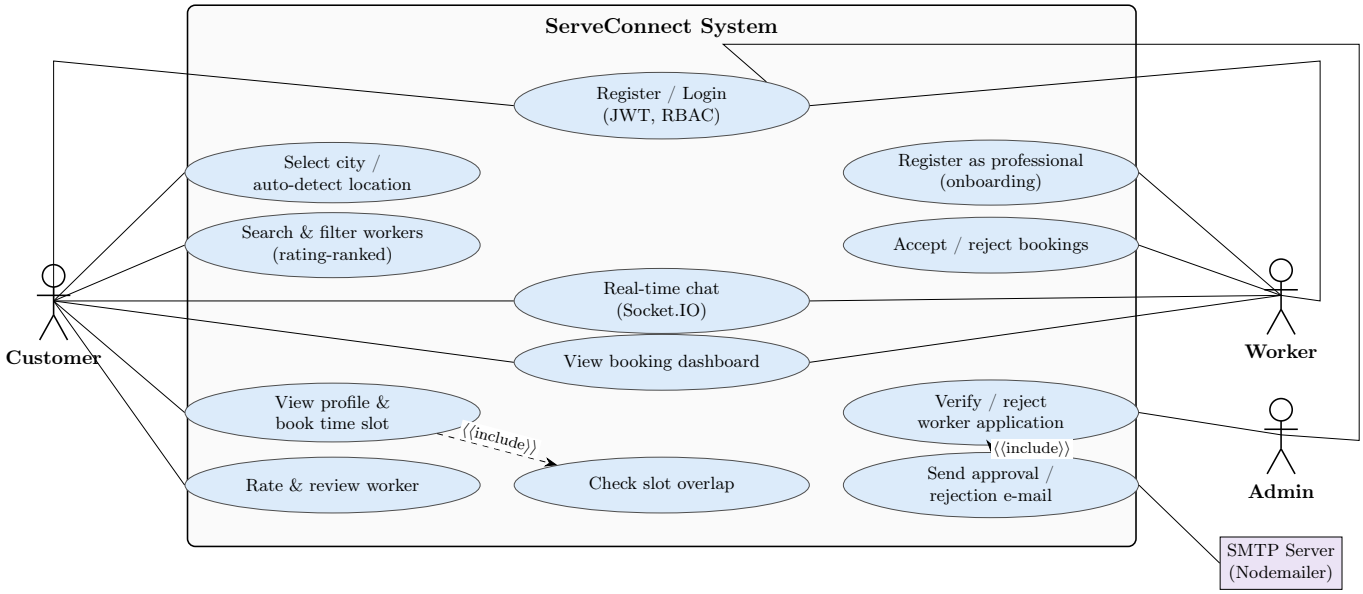


Figure 2: Use case diagram of ServeConnect for the Customer, Worker (professional) and Admin actors.

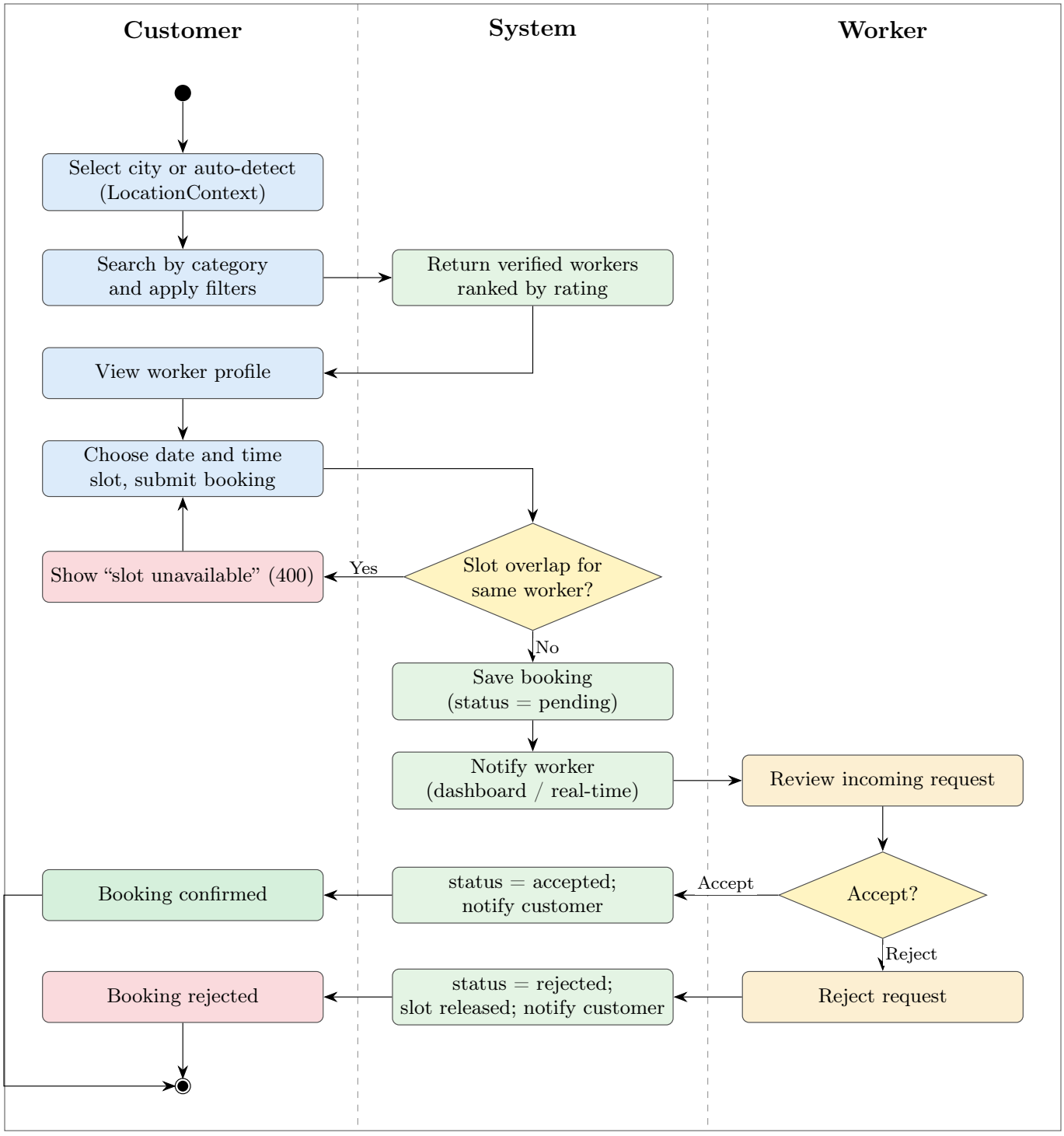


Figure 3: Activity diagram — customer booking flow with double-booking prevention.

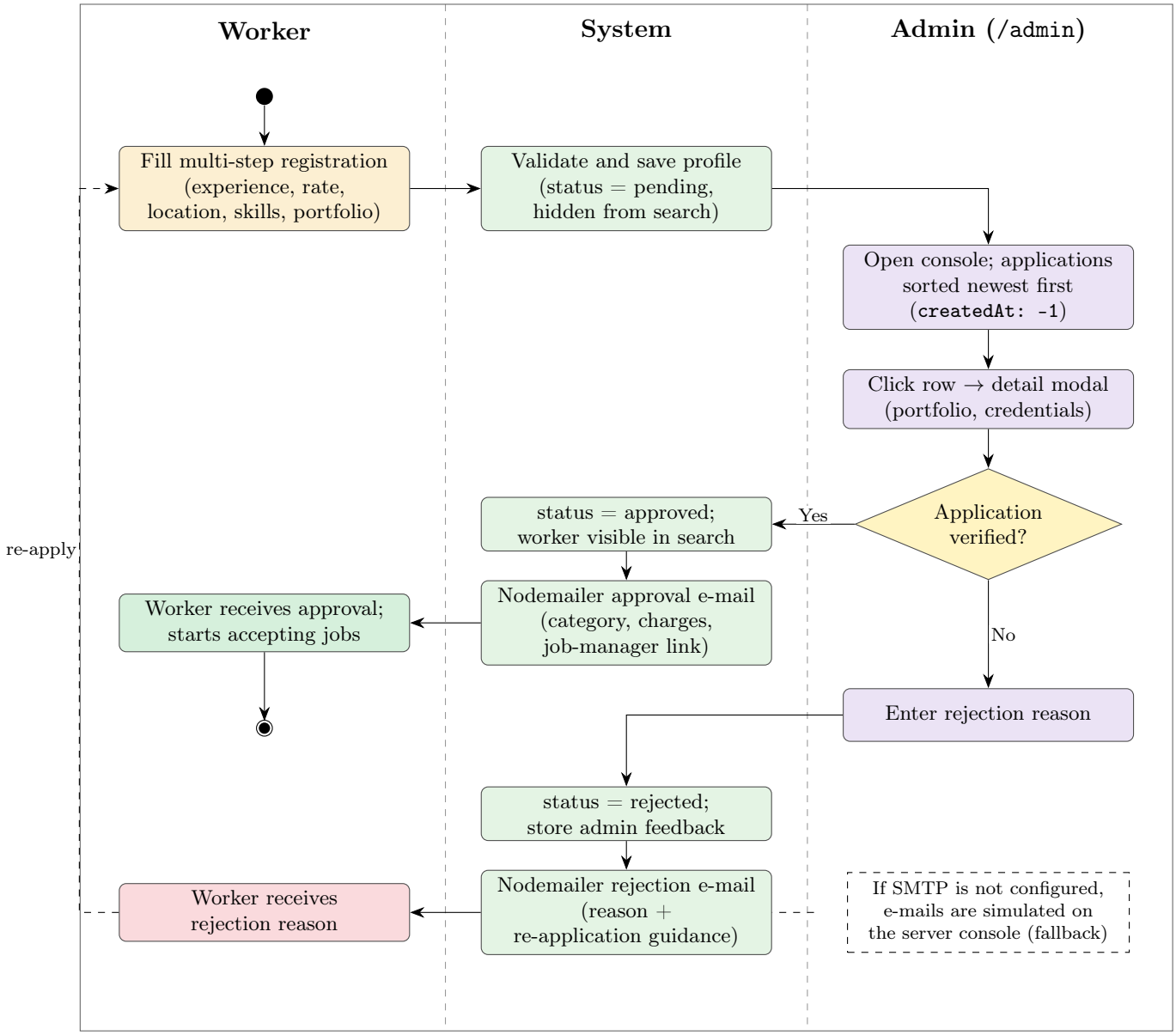


Figure 4: Activity diagram — admin worker-verification pipeline with automated approval and rejection e-mails.

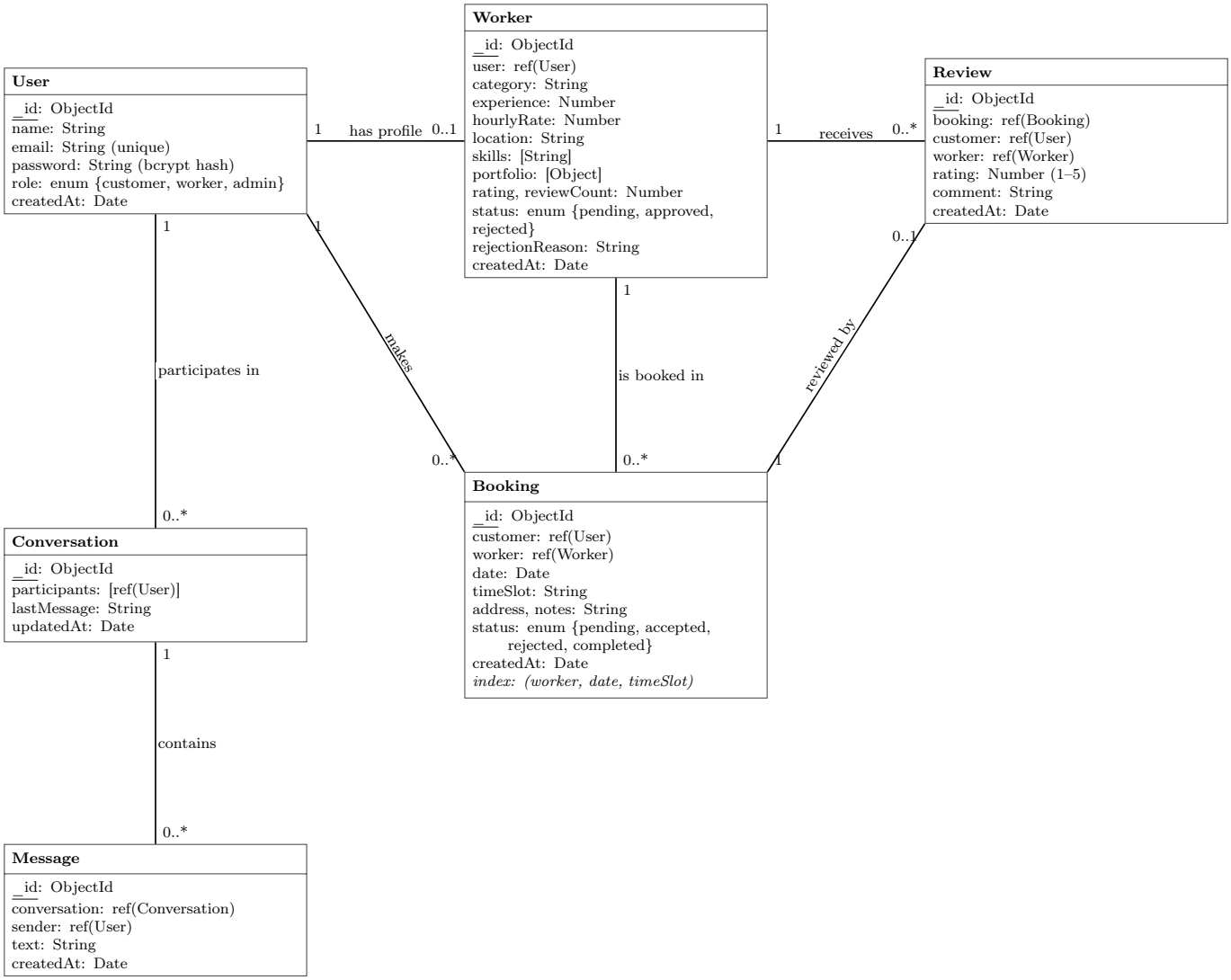


Figure 5: Class / entity-relationship diagram of the MongoDB (Mongoose) data model. Attribute lists are indicative; adjust to the final schemas.

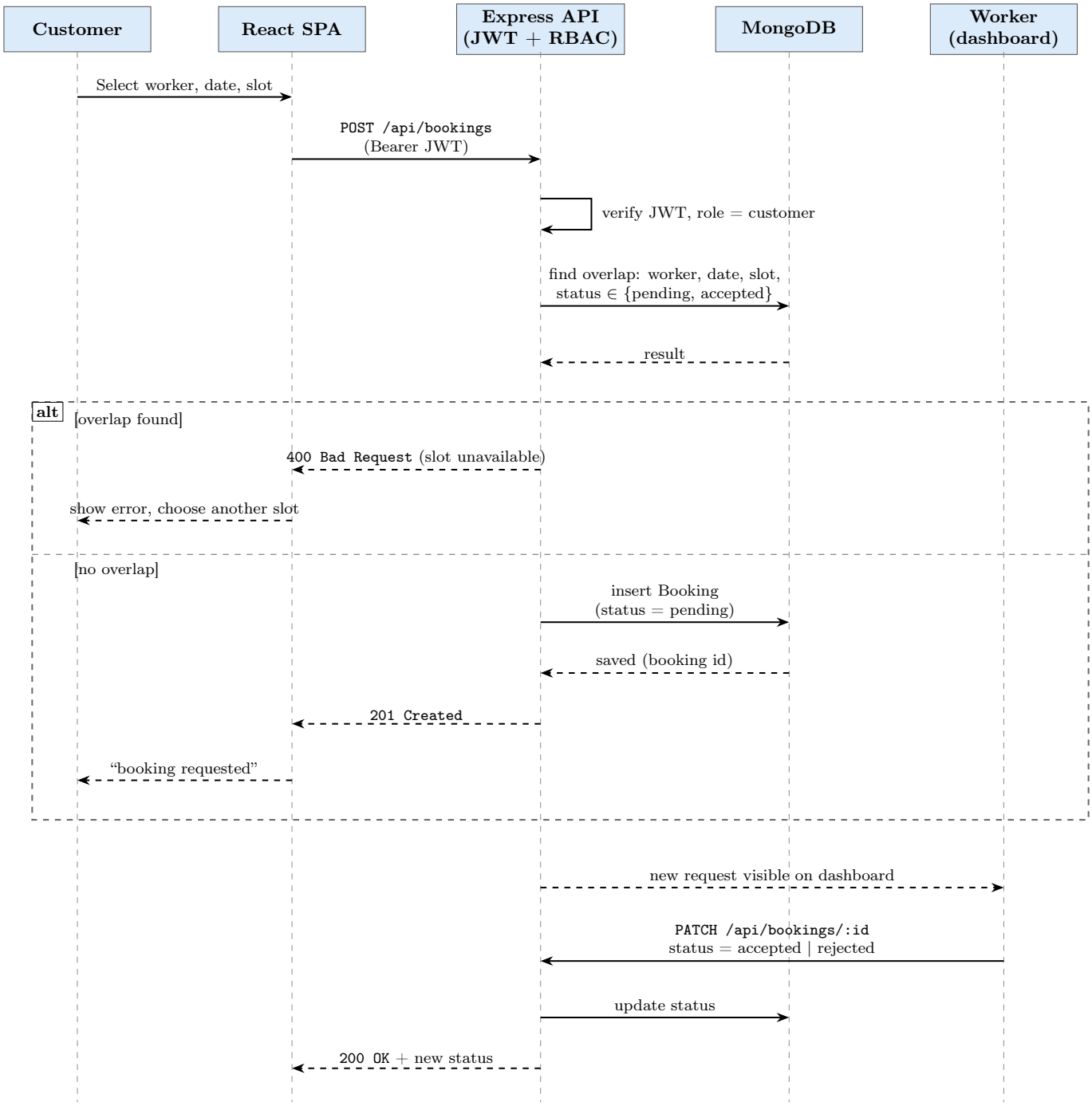


Figure 6: Sequence diagram — booking creation, conflict detection and worker response.

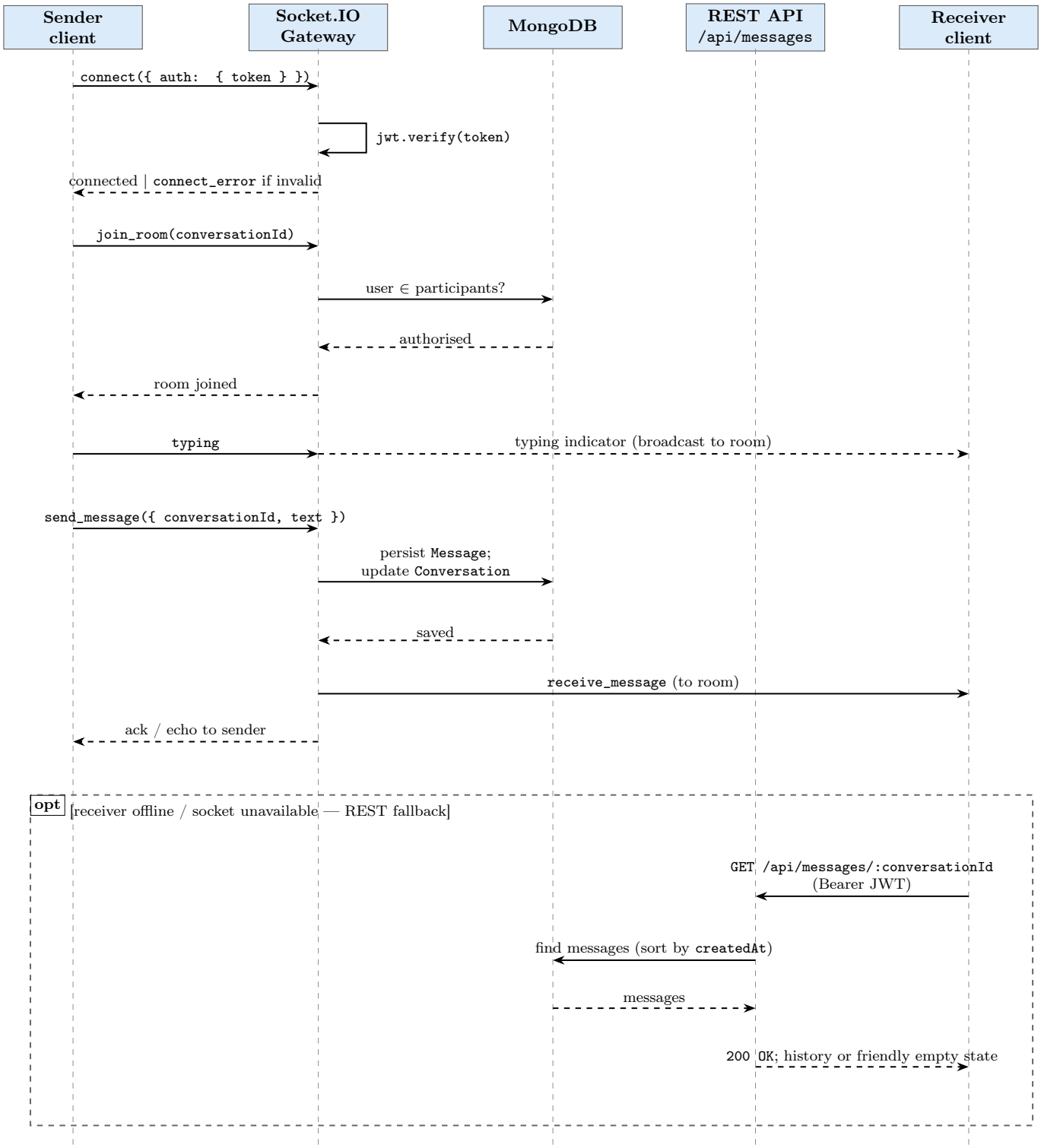


Figure 7: Sequence diagram — Socket.IO event lifecycle (authenticated handshake, room join, messaging, typing) with REST fallback.

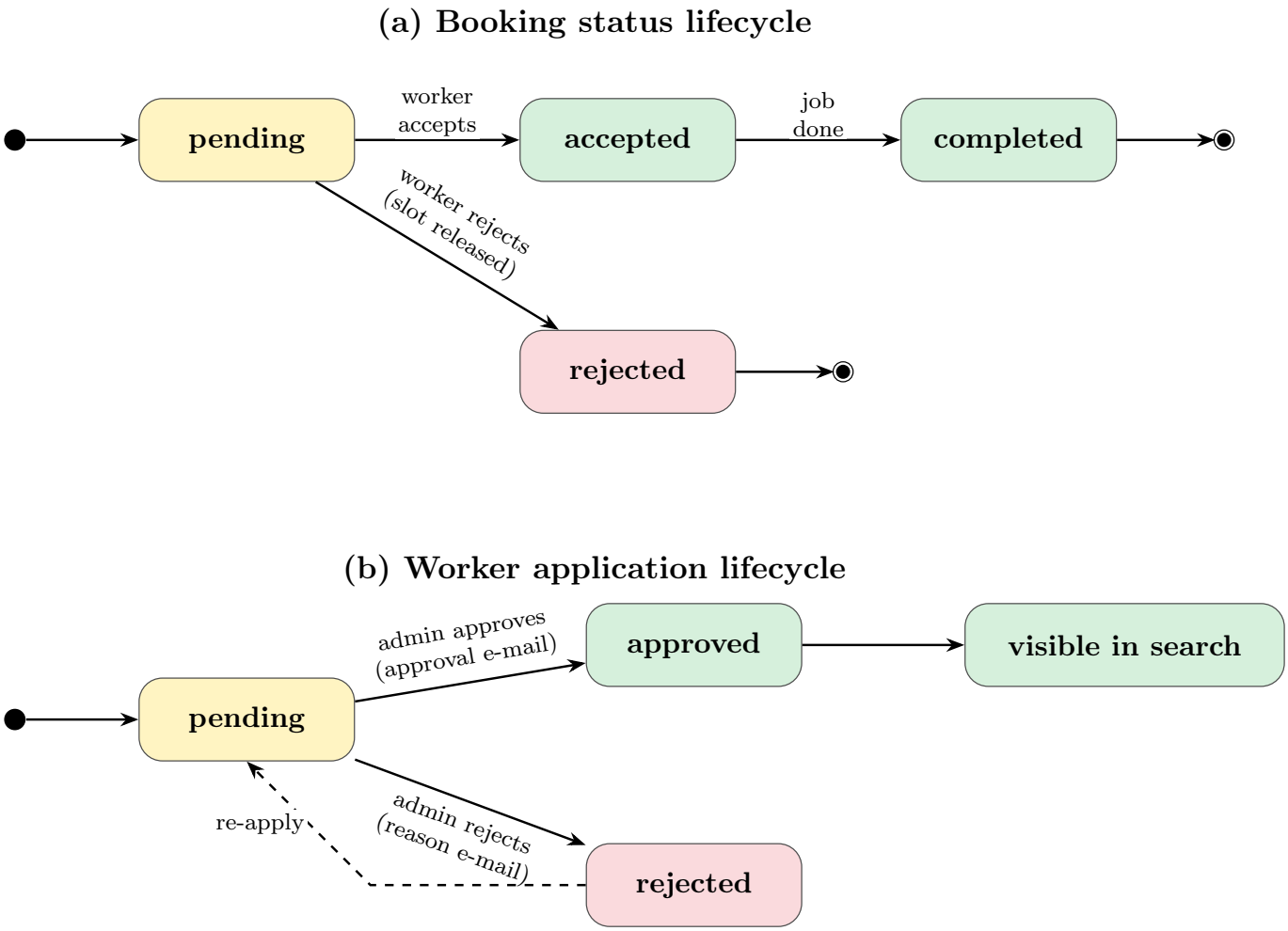


Figure 8: State diagrams for the booking status machine and the worker-verification lifecycle.

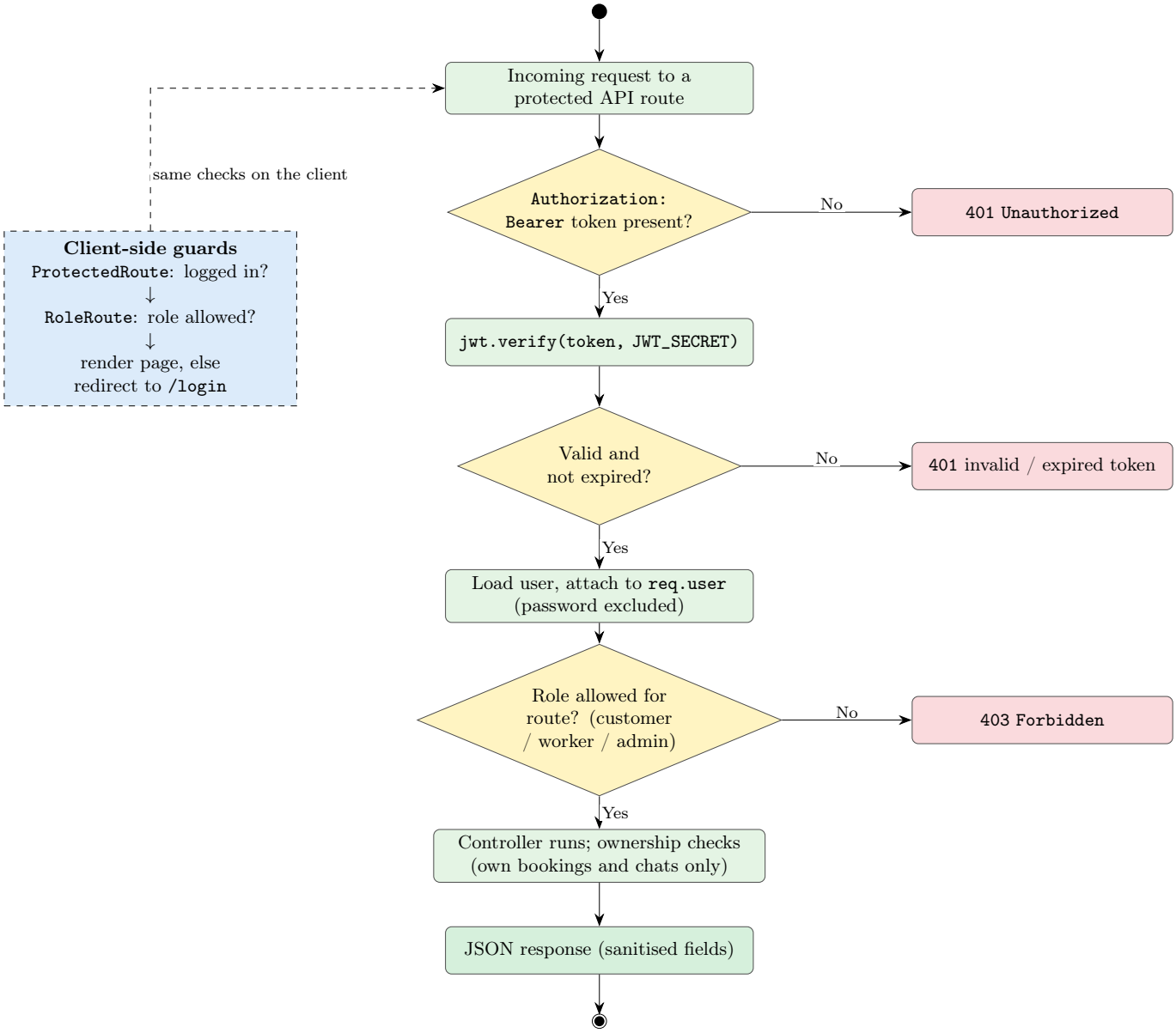


Figure 9: Stateless JWT authentication and role-based access control (RBAC) request pipeline.

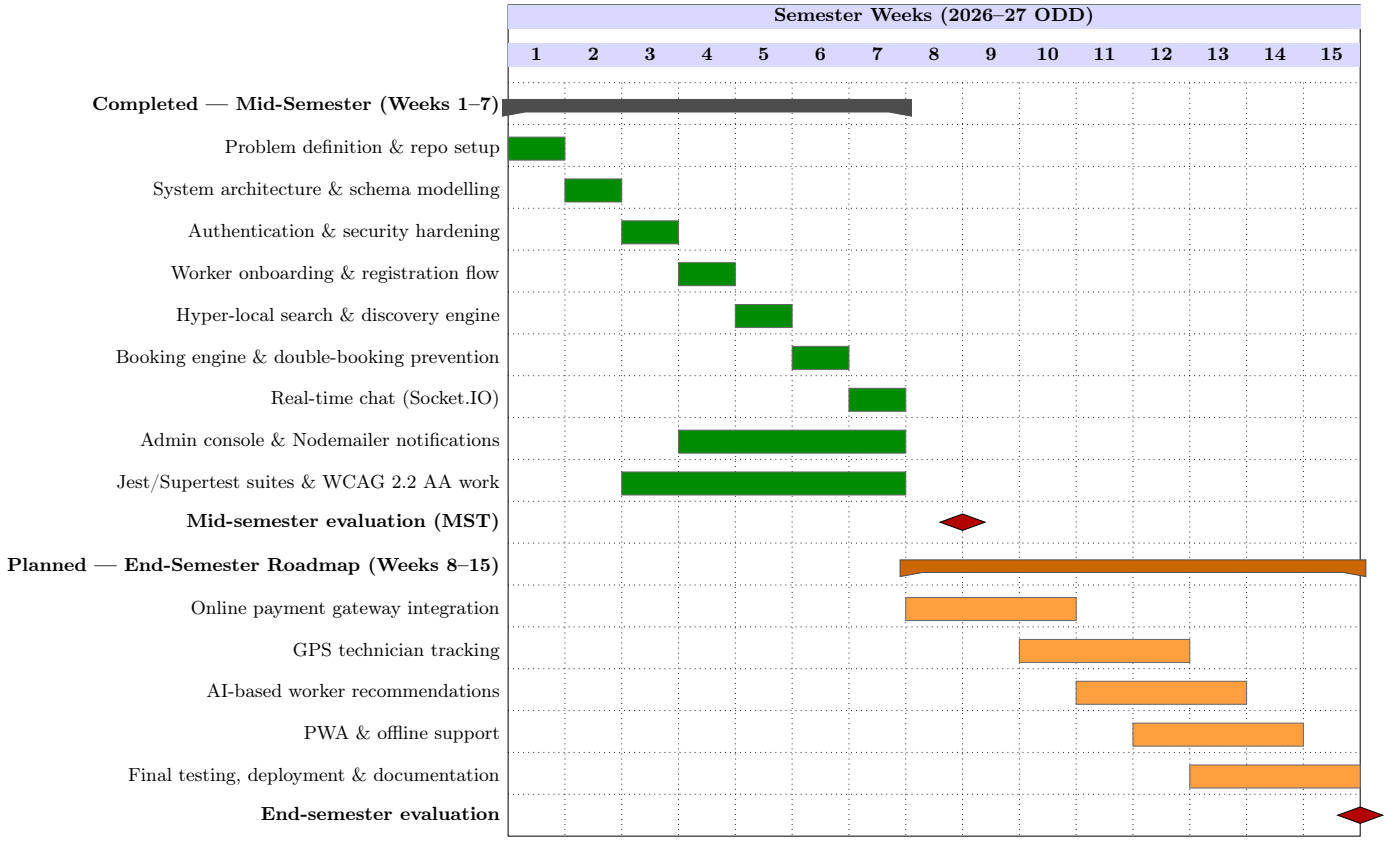


Figure 10: Project Gantt chart: completed work (green, Weeks 1–7) and planned end-semester roadmap (orange, Weeks 8–15). Adjust week numbers to your department calendar.